

JAVA CRUD

PART 2



Java[™]

@java_in.depth

CRUD is not just database operations. In real applications, it represents how your backend API interacts with data through proper layers.

Create, Read, Update, Delete operations are exposed via REST endpoints and handled through controllers, services, and repositories in Java backend systems.

1. PAGINATION & SORTING (MUST IN APIS)

Fetching all data at once is not scalable.
Real APIs always use pagination.

```
Page<User> users =  
userRepository.findAll(PageRequest.of(0, 10));
```

Pagination improves performance and
reduces load on database and network.

2. DTO MAPPING (CLEAN API DESIGN)

Never expose entities directly in APIs. Use DTOs to control what data is sent.

```
UserDto dto = modelMapper.map(user, UserDto.class);
```

This keeps your API secure, flexible, and decoupled from database structure.

3. VALIDATION (PRODUCTION LEVEL)

Validation prevents bad data from entering your system.

```
@NotNull  
@Email  
private String email;
```

In Spring Boot, validation is handled using annotations and improves API reliability.

4. GLOBAL EXCEPTION HANDLING

Handling errors properly is a must in production APIs.

```
@RestControllerAdvice  
public class GlobalExceptionHandler {  
}
```

Centralized exception handling ensures consistent error responses.

5. TRANSACTION MANAGEMENT

Transactions ensure data consistency.

```
@Transactional  
public void transferMoney() {  
}
```

If one operation fails, everything rolls back, preventing inconsistent data.

6. CACHING (PERFORMANCE BOOST)

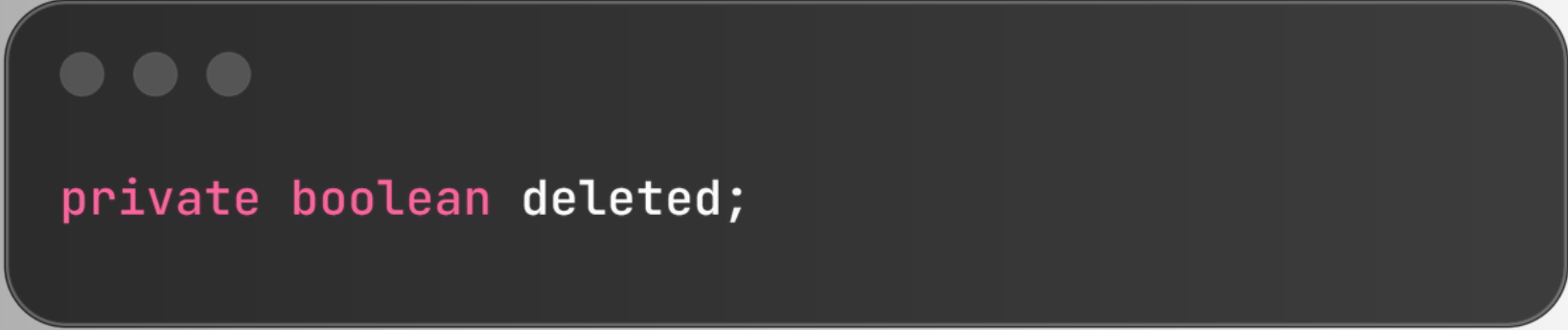
Caching reduces database load and improves response time.

```
@Cacheable("users")  
public User getUser(Long id) { }
```

Used heavily in high-performance systems.

7. SOFT DELETE VS HARD DELETE

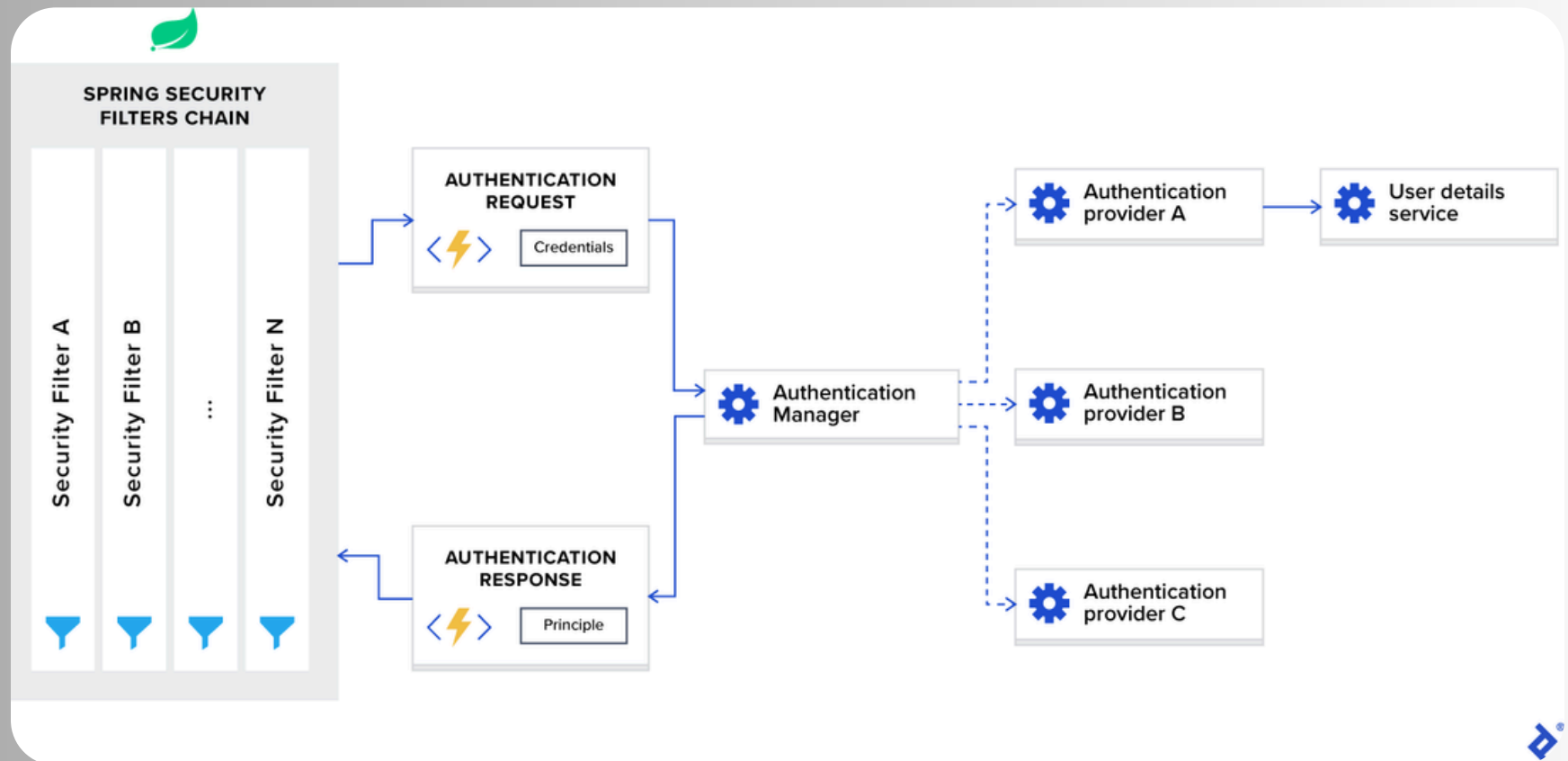
Instead of deleting data permanently, many systems use soft delete.



```
private boolean deleted;
```

This helps in auditing, recovery, and compliance.

8. API SECURITY BASICS



CRUD APIs must be secured.

Common approaches:

- JWT Authentication
- Role-based Authorization

Security is often asked in backend interviews.

8. WHAT MAKES YOU STAND OUT

Advanced CRUD is about thinking like a backend engineer.

It includes performance, scalability, security, and clean architecture—not just database operations.

This is what separates beginners from strong developers.

WANT PART 3

LET ME KNOW IN COMMENTS!

Save this post and apply these in your projects.